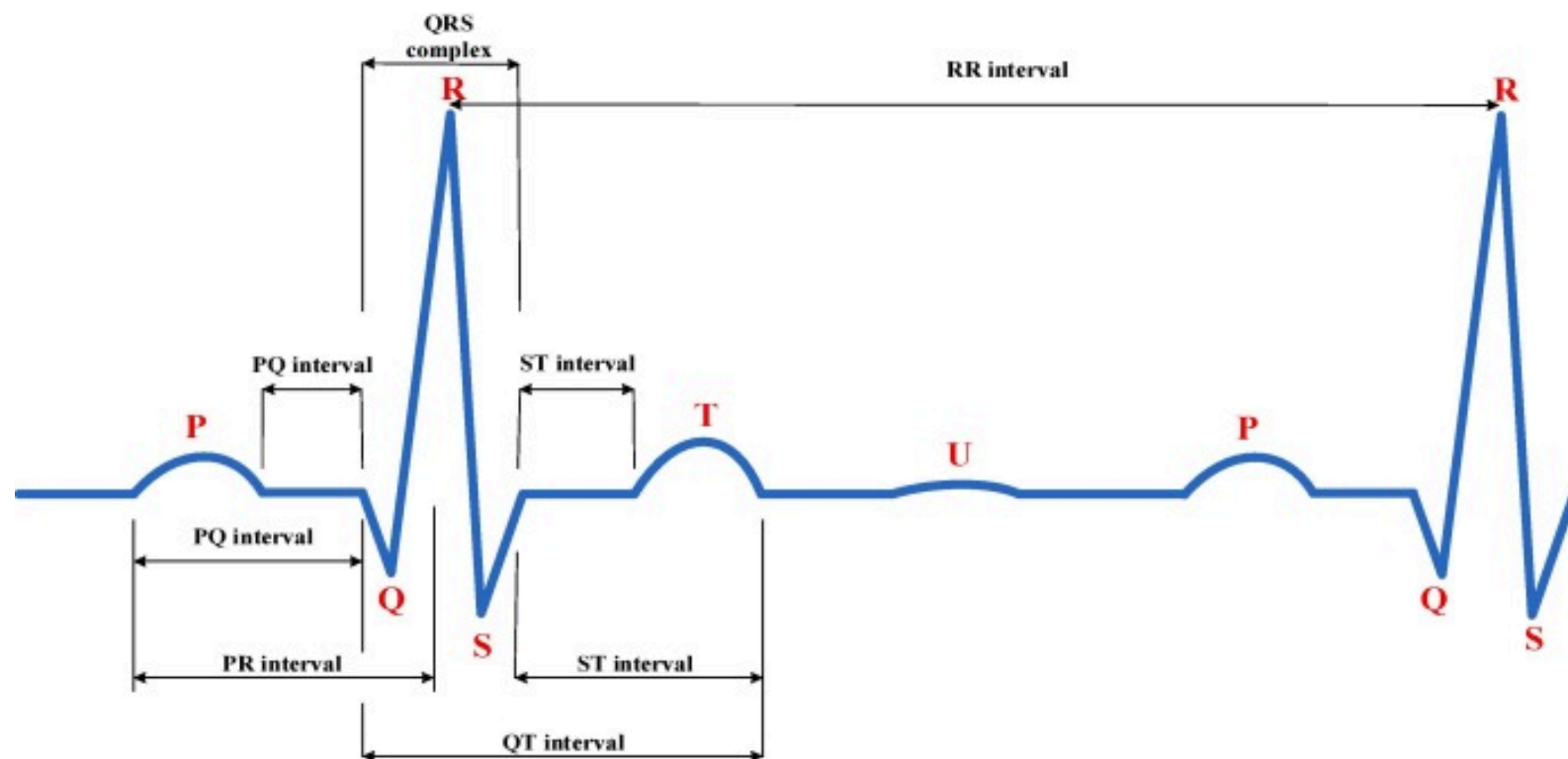
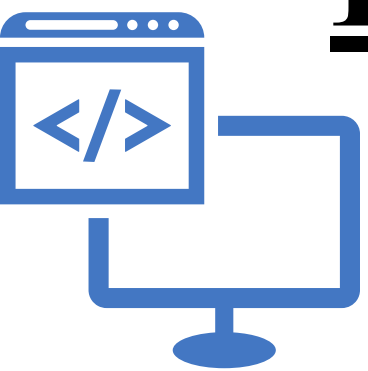


Fast & Fouriers



ECG Arrhythmia Detection using Fourier Transform and Hybrid CNN-Vision Transformer Model

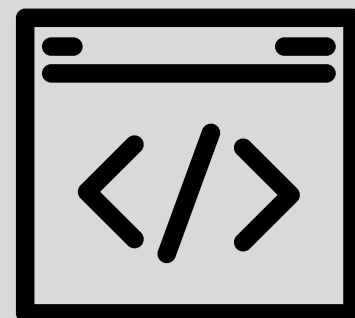
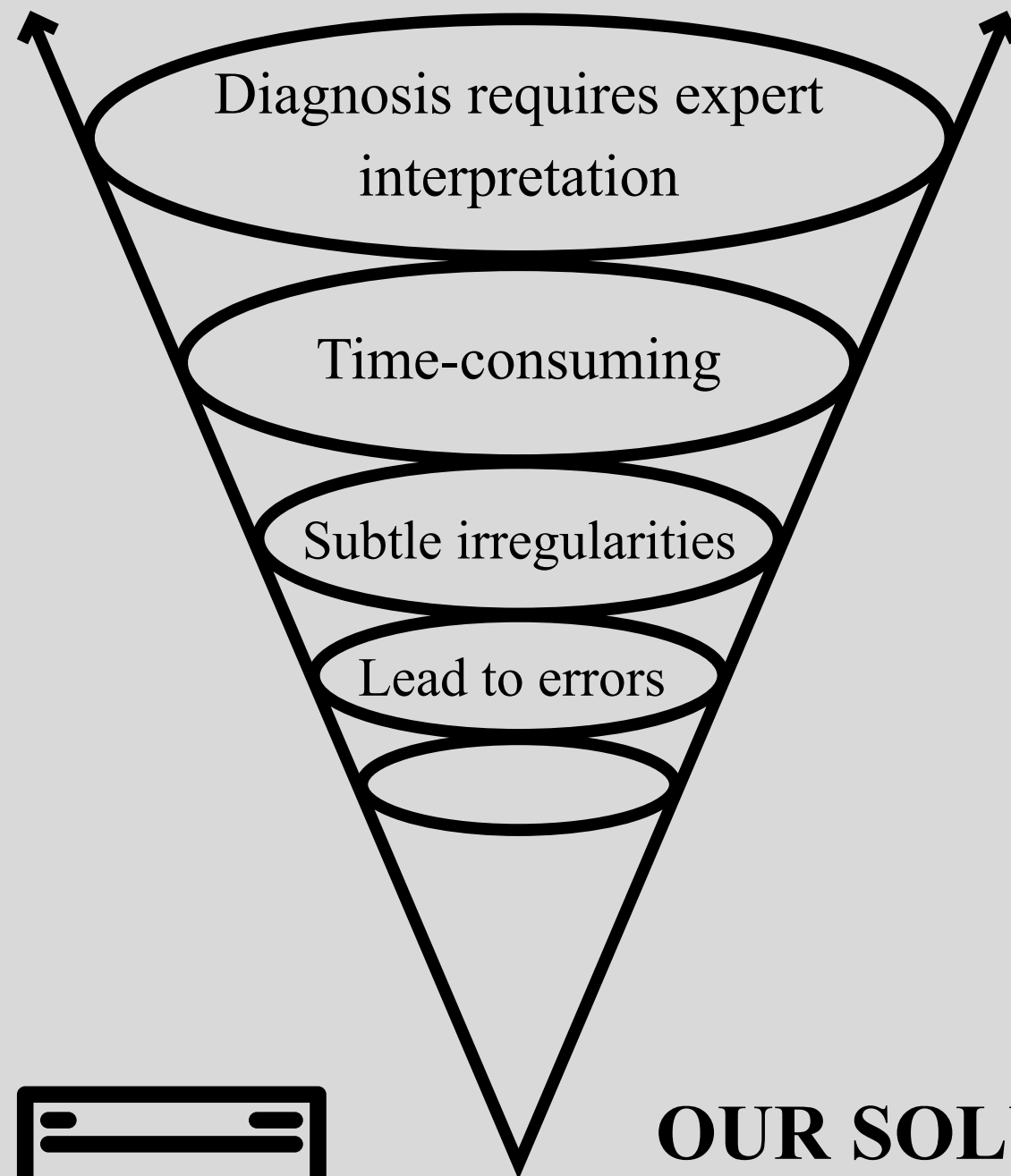


Preeti Shanbhag - 20230901026
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Mentored by: Dr. Prabir Kumar Das
TA: Mr. Nitish Sawant & Mr. Jay Kumar

Introduction - The Problem & Our Solution

Cardiac arrhythmia is difficult to detect manually because of the following reasons:



OUR SOLUTION

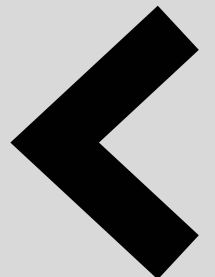
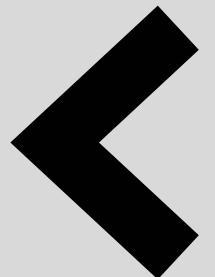
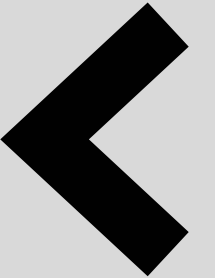


To develop a Hybrid CNN + Transformer model for automatic detection and classification of arrhythmias.

To analyze and preprocess ECG signals using Fourier Transform for better feature extraction and noise reduction.

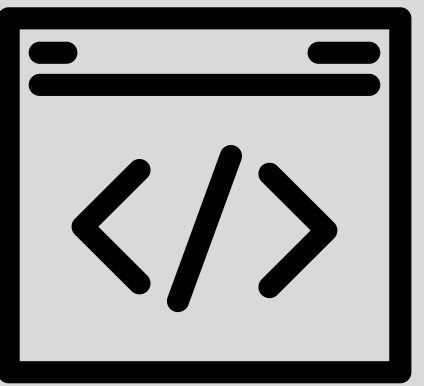
To improve early cardiac diagnosis by creating a fast, accurate, and intelligent support system for healthcare.

←
OUR OBJECTIVES:



An automated AI-based system that does all of this.

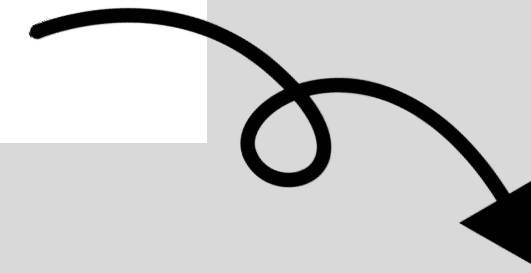
Dataset & Research Details



Integrated dataset:

ECG Heartbeat Categorization Dataset

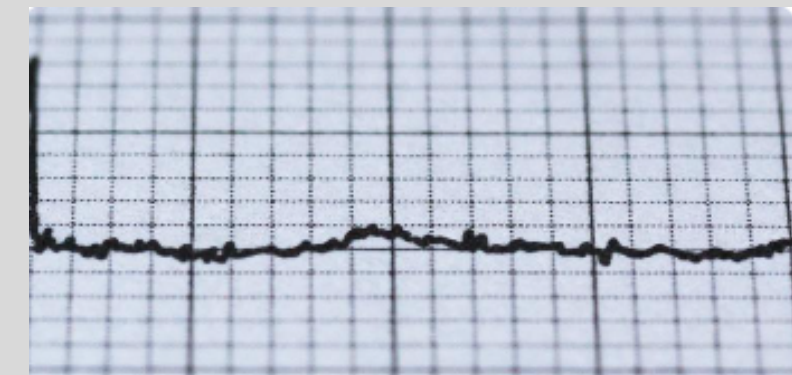
Segmented and Preprocessed ECG Signals for Heartbeat Classification



- Contains 109,446 ECG heartbeat recordings
- Derived from MIT-BIH Arrhythmia Database and PTB Diagnostic ECG Database
- Includes signals from multiple patients with normal and abnormal cardiac conditions
- Each heartbeat sample contains 188 numerical signal points
- Sampling frequency: 125 Hz
- Total Classes: 5 heartbeat categories
- Classes include: Normal, Supraventricular, Ventricular, Fusion, Unknown beats
- Data provided in CSV format for machine learning and deep learning analysis

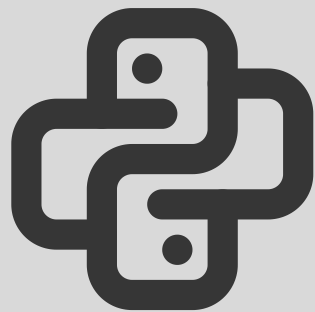
Link to our dataset:

<https://www.kaggle.com/datasets/shayanfazeeli/heartbeat/data>



What is the difference between the MIT-BIH Arrhythmia Database and the PTB Diagnostic ECG Database?

- MIT-BIH focuses on heartbeat rhythm abnormalities (arrhythmias).
- PTB focuses on overall heart disease diagnosis.



Existing Literature & Our Work

Binary Arrhythmia Classification

(Normal vs Abnormal heartbeat detection: ECG signals)

<https://link.springer.com/article/10.1007/s43926-025-00131-7>

Hybrid CNN + Transformer Architecture

(Combines local feature extraction with sequence learning)

Multi-Class ECG Beat Classification

(Detection of multiple heartbeat classes using the MIT-BIH dataset)

<https://arxiv.org/abs/2303.03660>

Fourier Transform Based Analysis

(Time domain to frequency domain conversion for better features)

Use of Individual Deep Learning Models

(CNN, LSTM, RNN for ECG signal analysis)

<https://arxiv.org/abs/2209.00988>

Improved ECG Signal Understanding

(Detection of hidden frequency patterns and noise removal)

Hybrid Deep Learning Architectures

(CNN-BLSTM / CNN-LSTM for better ECG classification)

<https://www.nature.com/articles/s41598-025-17671-1>

Multi-Class Arrhythmia Detection

(Classifies different heartbeat categories accurately)

Advanced Preprocessing Techniques

(Filtering, segmentation, normalization before model training)

<https://www.sciencedirect.com/science/article/pii/S0893608023001211>

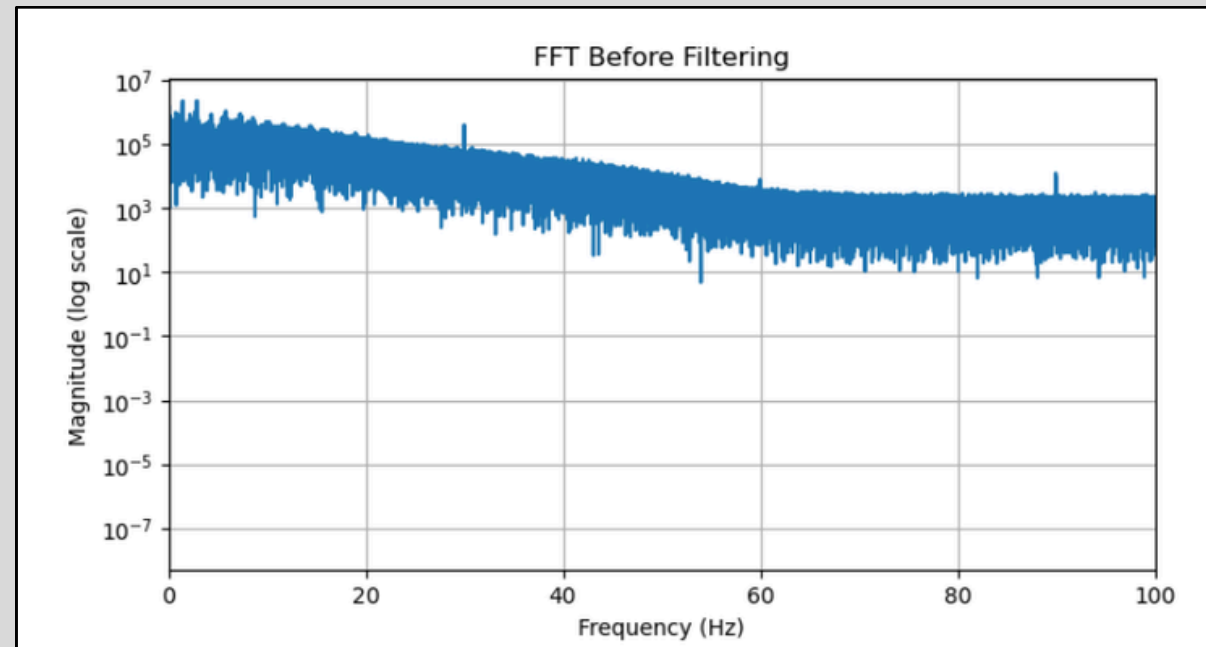
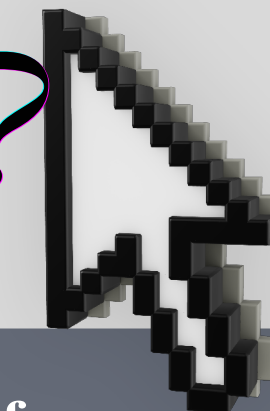
AI-Based Smart Cardiac Screening System

(Fast, automated, and efficient diagnosis)

Existing Literature

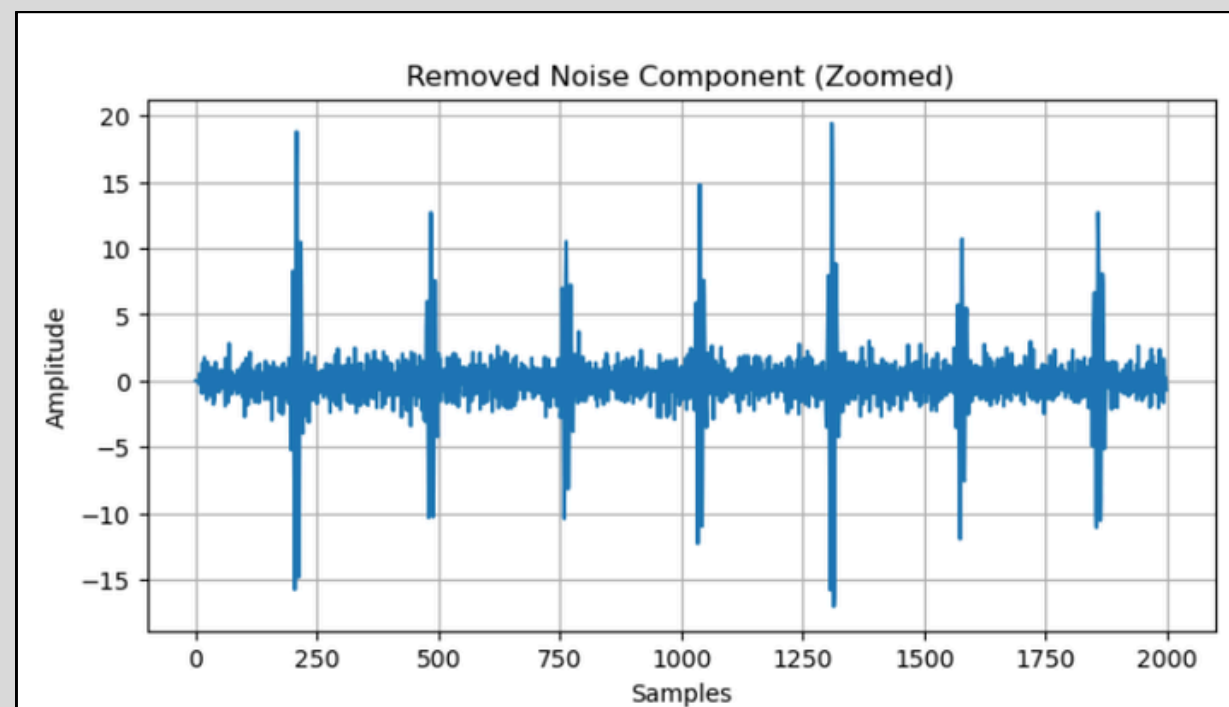
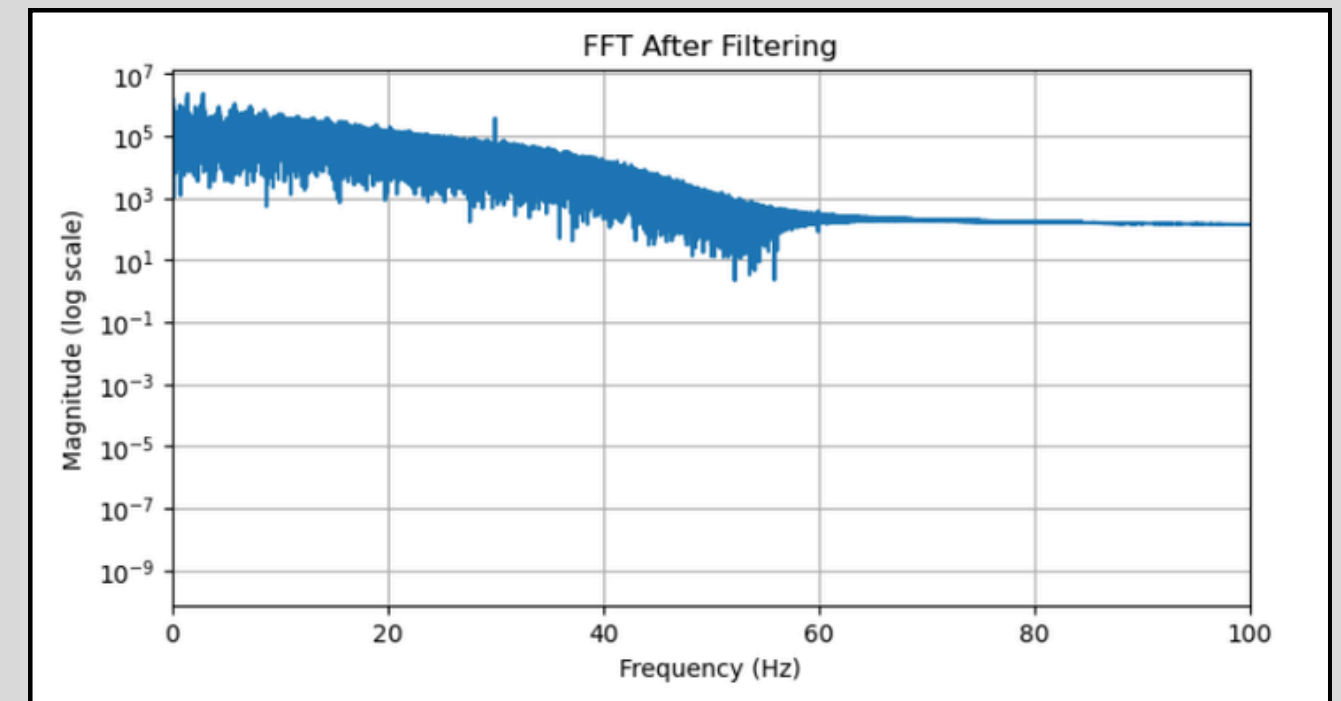
Our Work

Why Fourier Transform?



FFT of raw ECG shows wide frequency spread with low-frequency baseline drift and high-frequency noise, masking key cardiac signals.

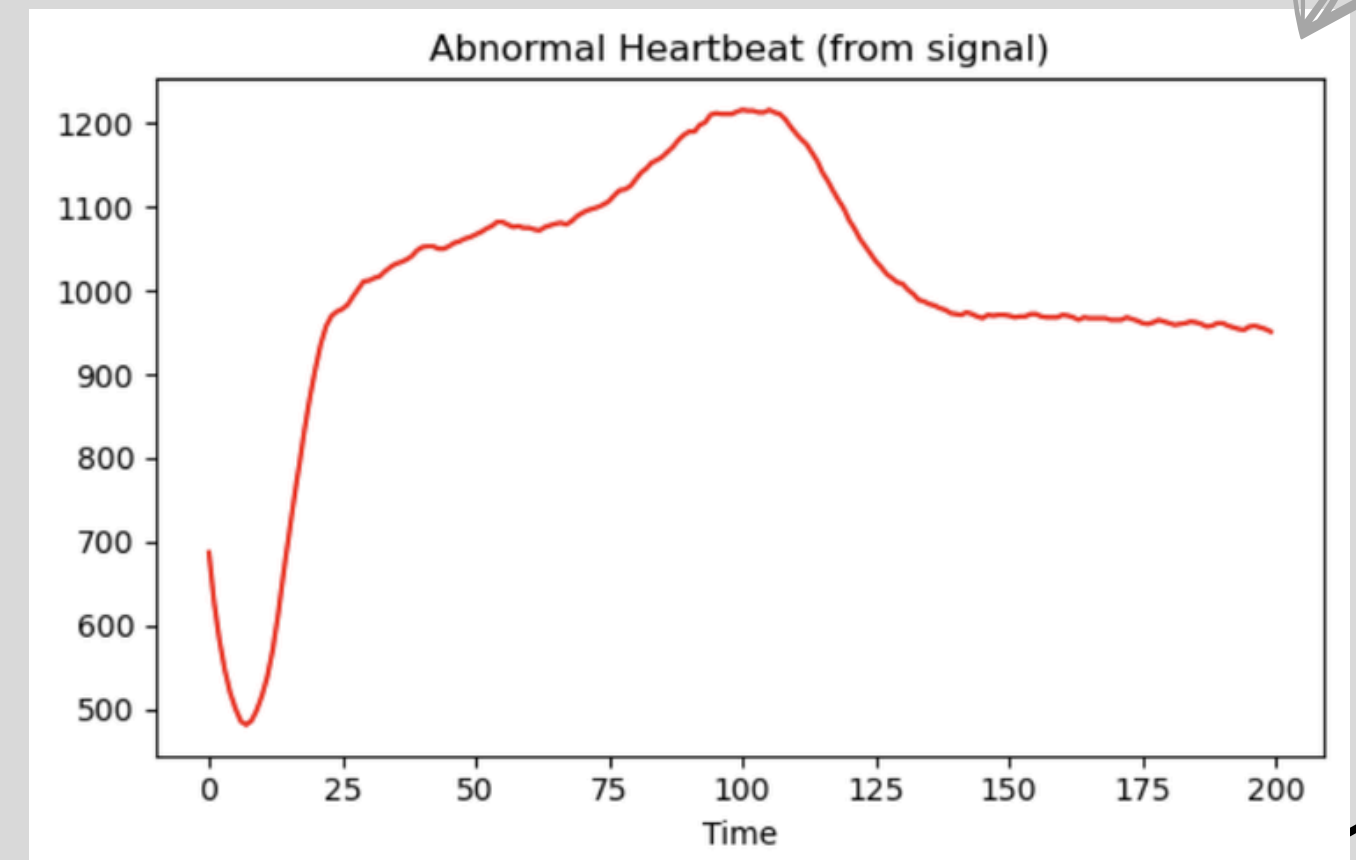
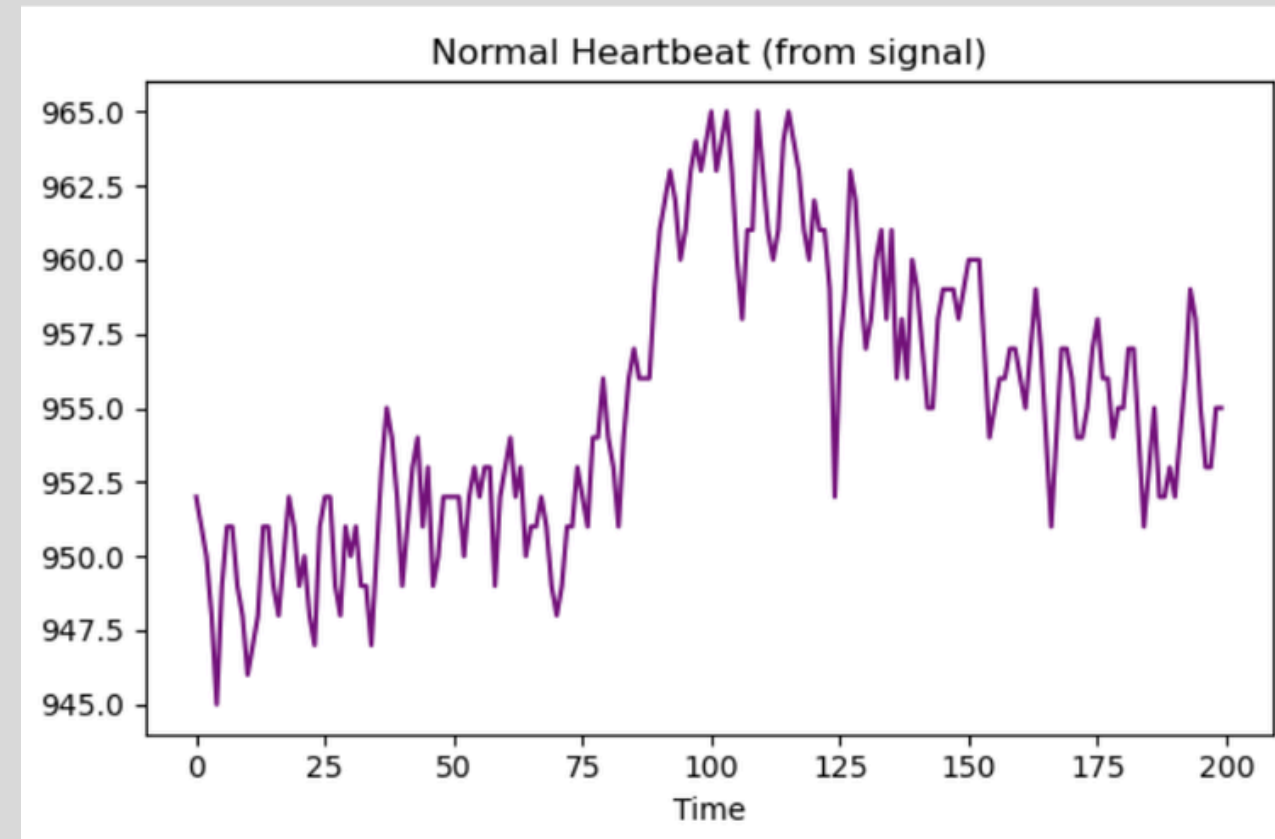
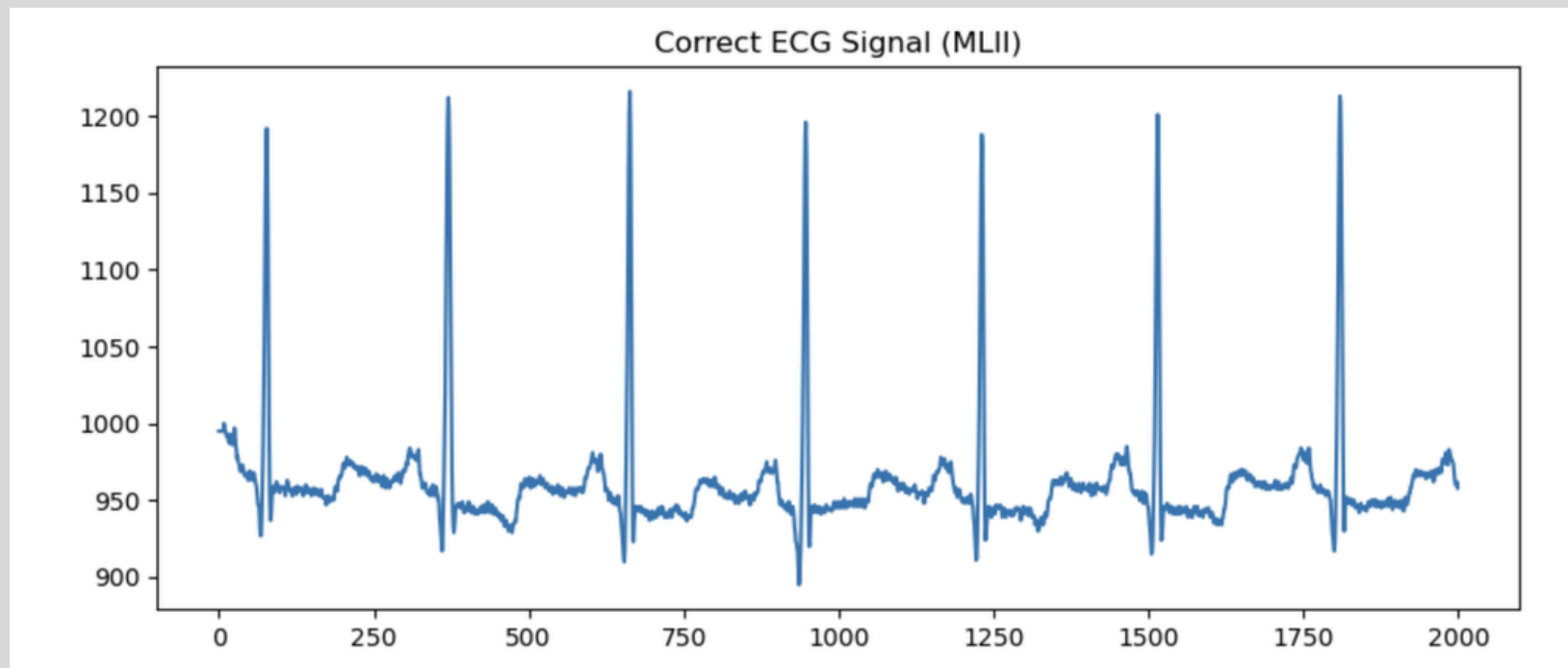
FFT after filtering shows reduced noise with energy concentrated in the relevant ECG frequency range, preserving key cardiac signals.



Removed noise shows isolated high-frequency artifacts and irregular fluctuations, confirming effective noise reduction.

FREQUENCY DOMAIN: NORMAL AND ABNORMAL HEARTBEAT

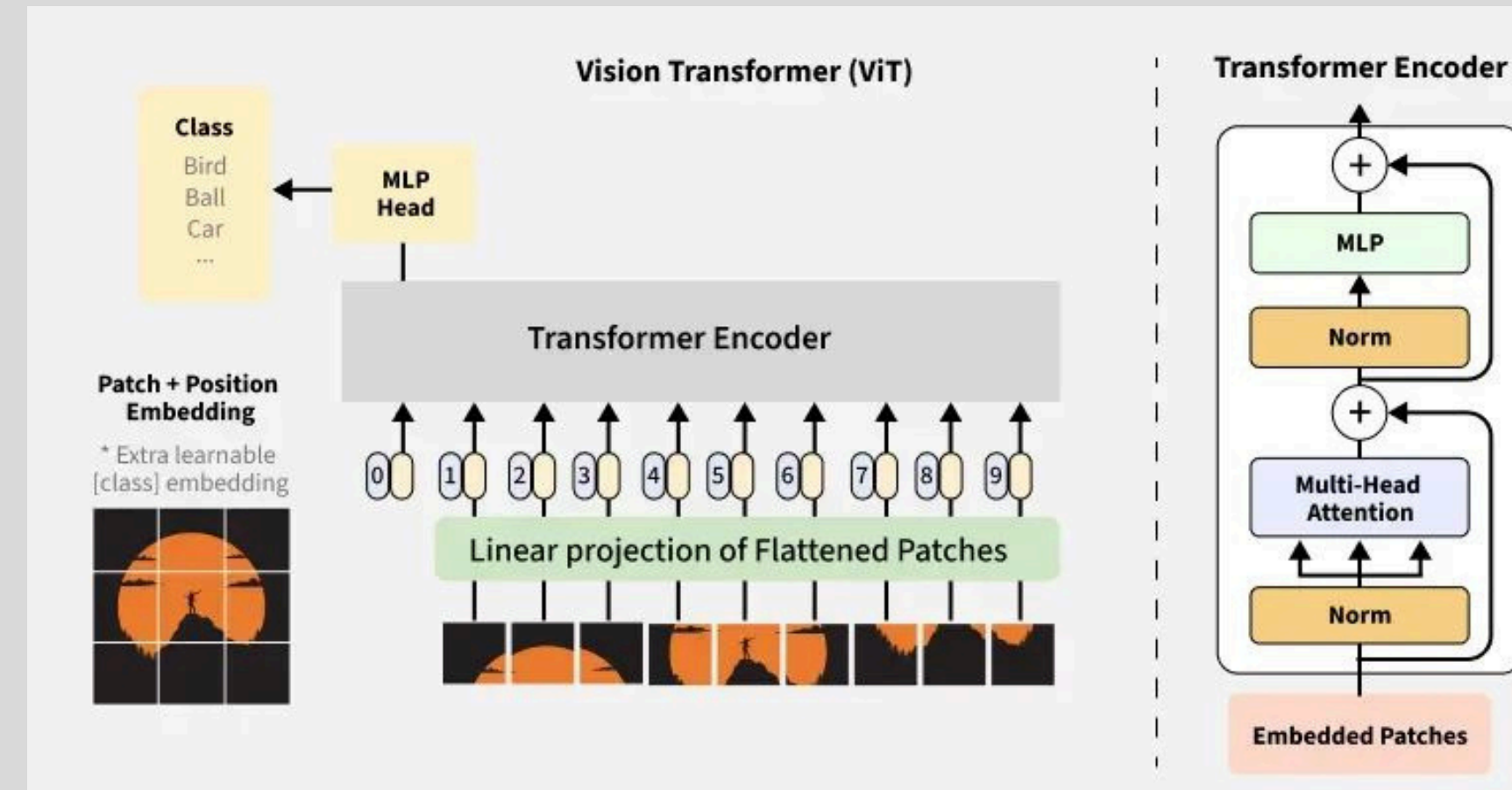
TIME DOMAIN



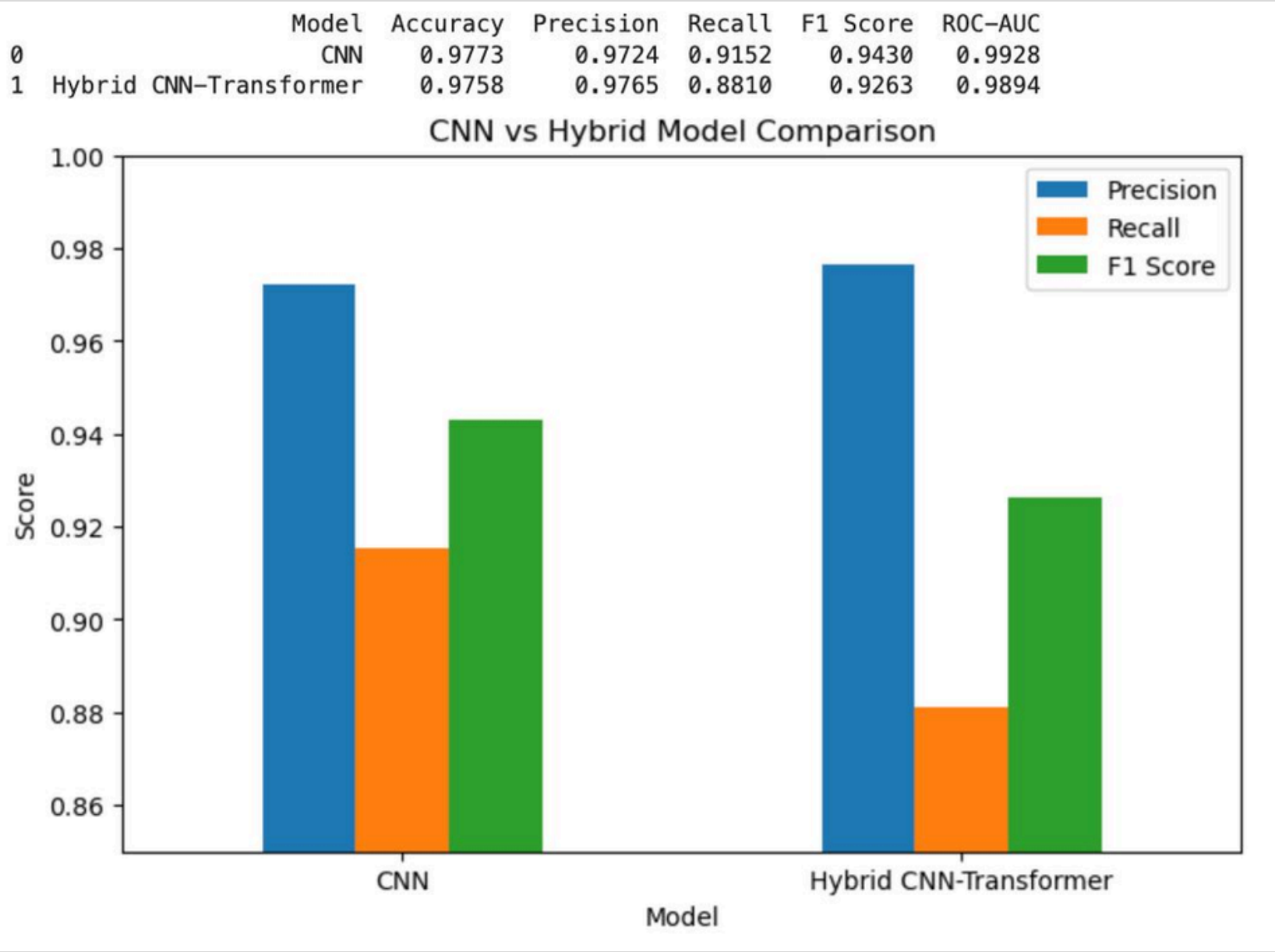
Time domain is converted
into Frequency Domain

WHY ITS NOVEL ?

- Most existing methods use CNN or CNN-LSTM, which mainly capture local waveform features
- Our model integrates Fourier Transform preprocessing to remove noise and enhance signal quality
- CNN is used to extract local ECG patterns like peaks and waveform shapes
- The ViT/Transformer introduces attention, capturing global dependencies across the entire signal
- It also helps the model focus on important regions of the ECG signal
- This combination provides both local and global feature understanding
- Results in better feature representation and more reliable classification compared to traditional methods

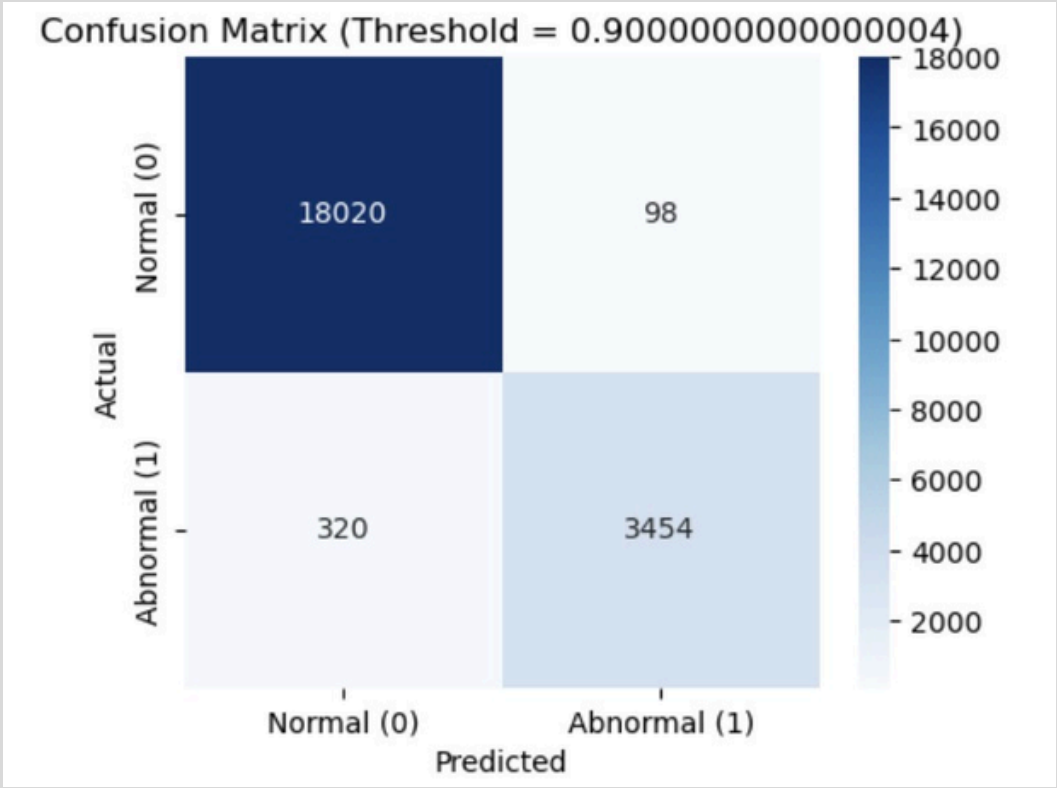


COMPARATIVE STUDY OF DIFFERENT MODALS



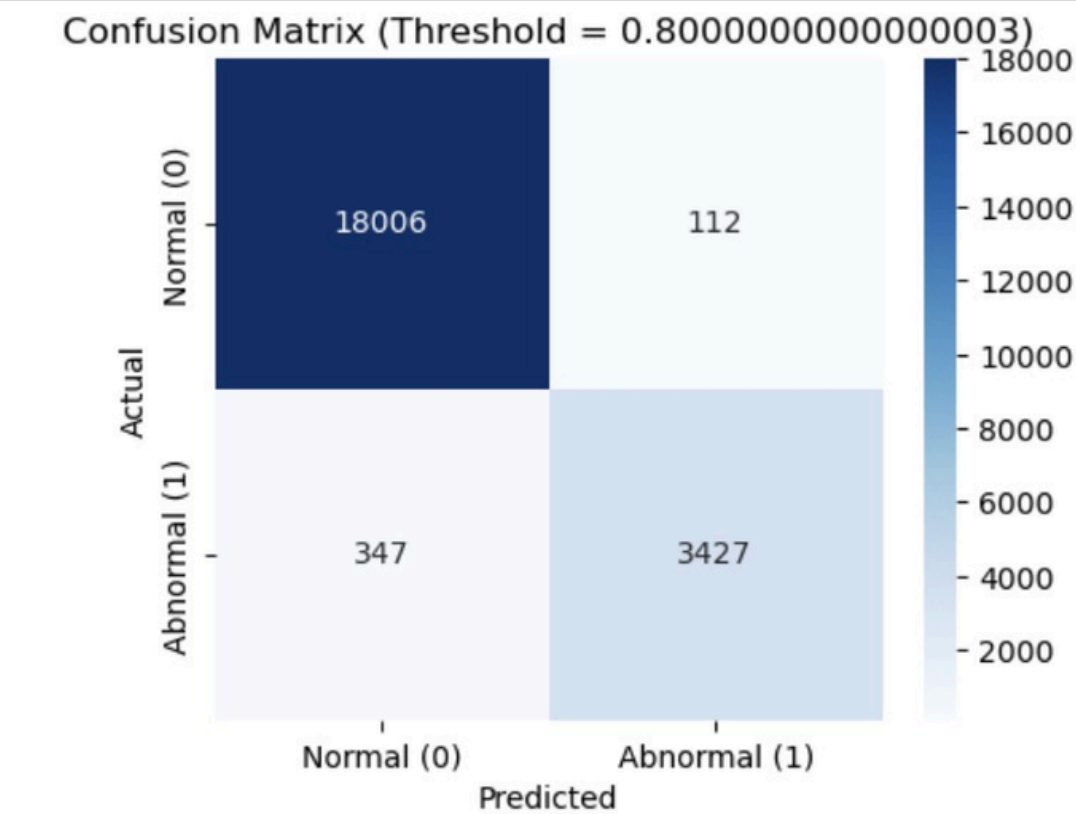
Hybrid model is more precise,
CNN is more sensitive

CNN



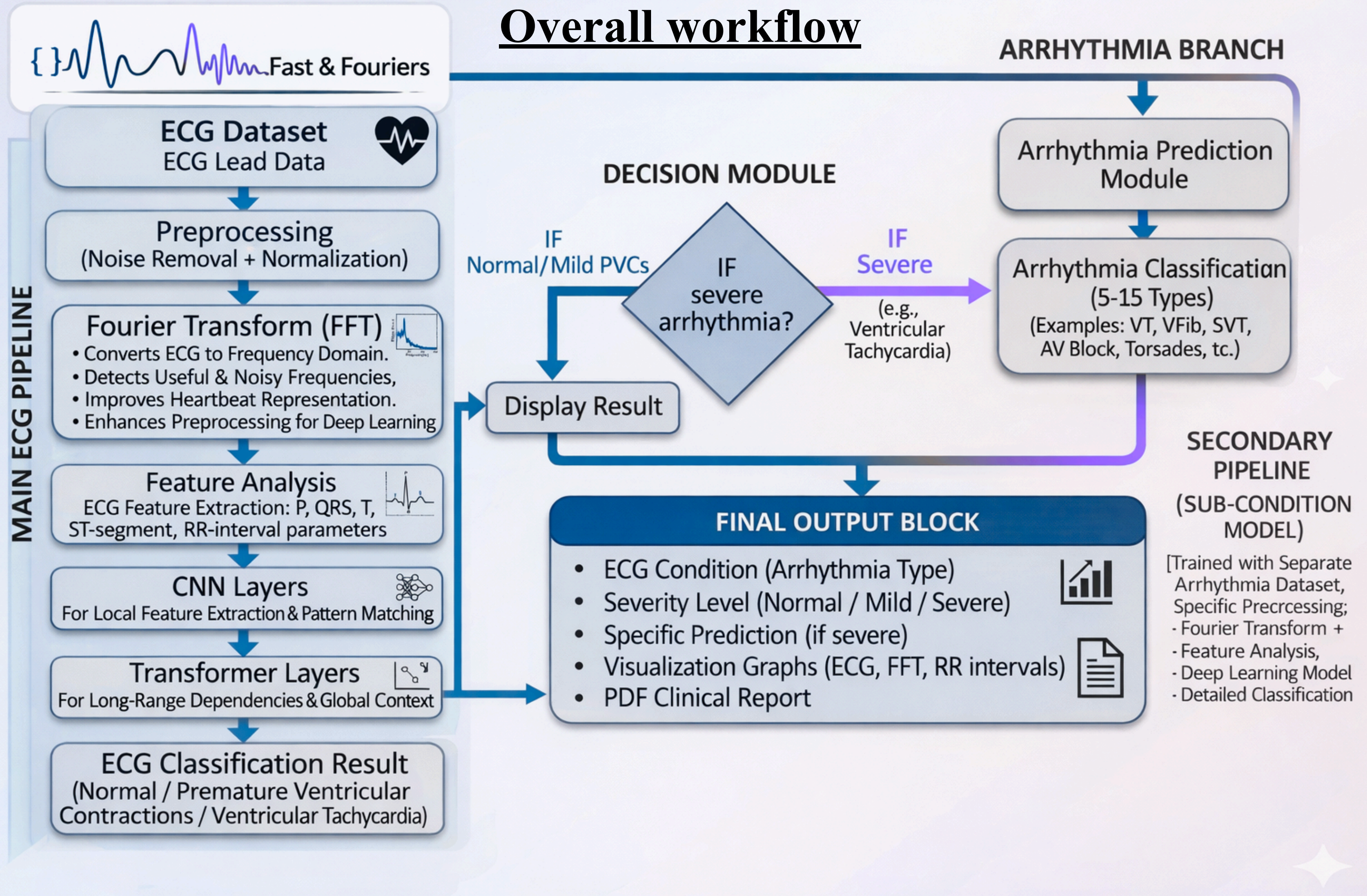
Accuracy : 97.7
Precision : 97.2

Hybrid CNN- Transformer



Accuracy : 97.6
Precision : 97.7

Overall workflow

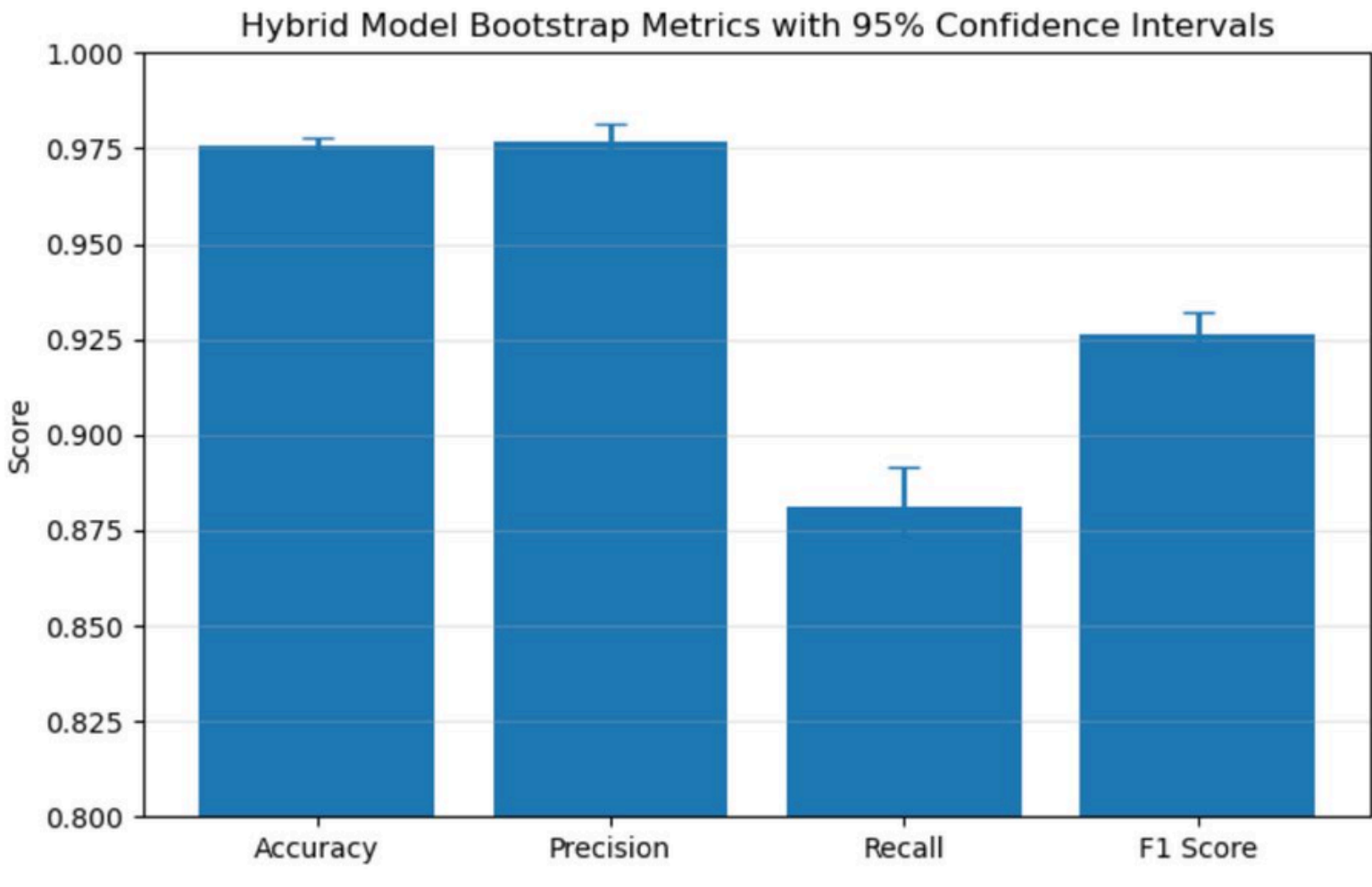


STATISTICAL VALIDATION

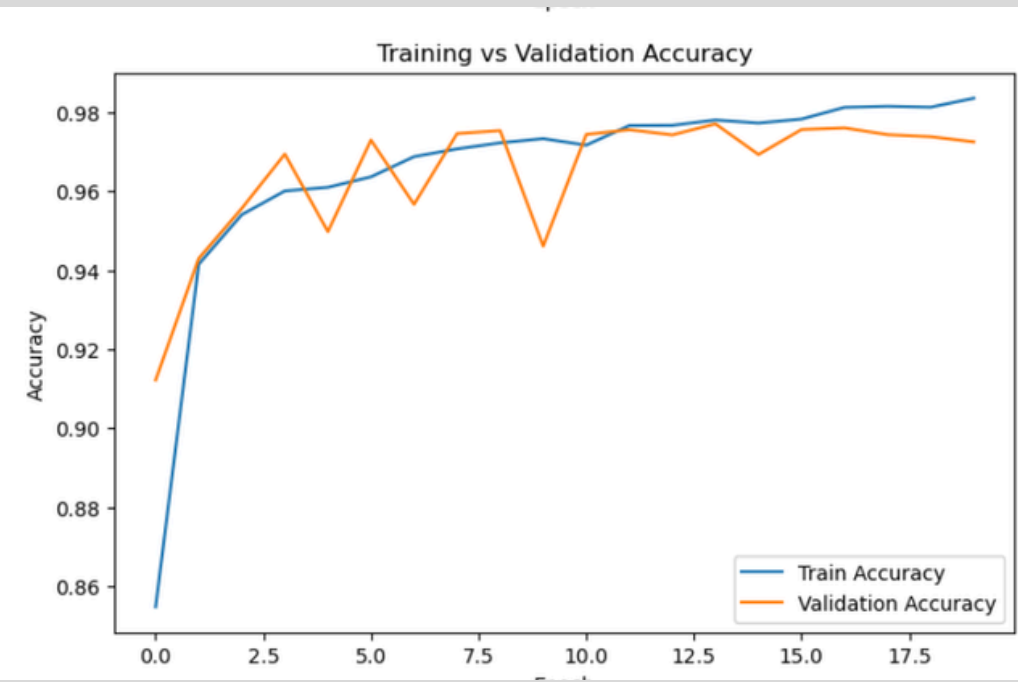
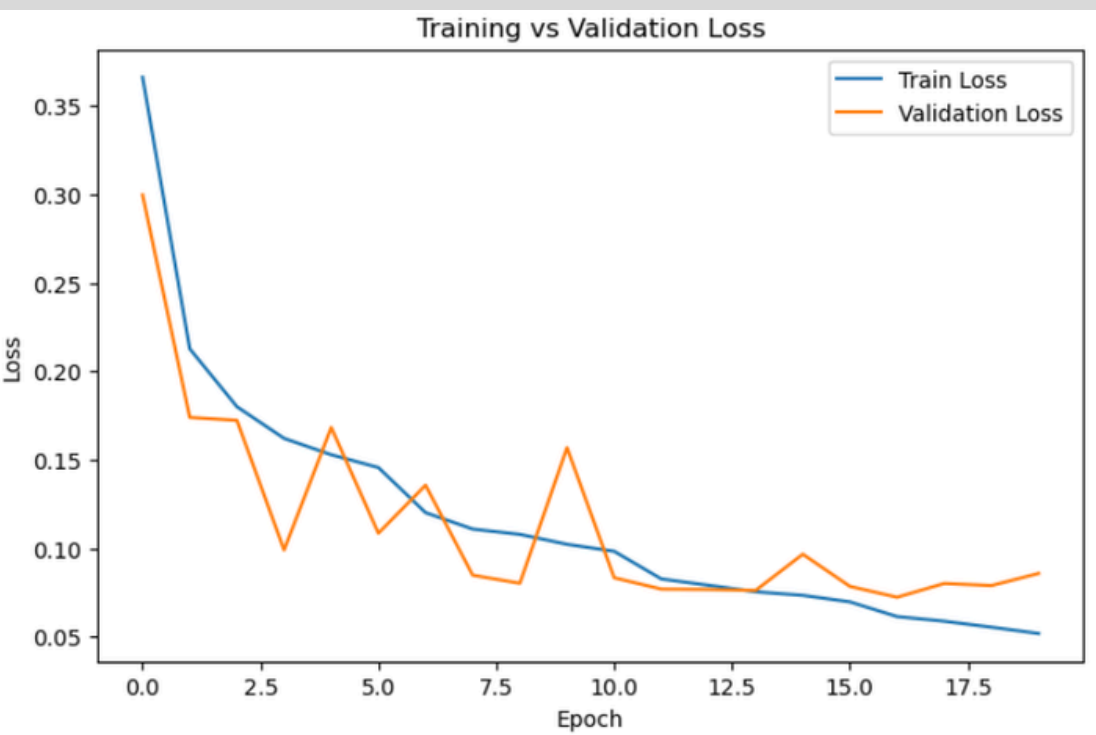
Results

Hybrid Model Bootstrap Results (95% CI):

Accuracy: 0.9758 (0.9738 – 0.9778)
Precision: 0.9765 (0.9713 – 0.9814)
Recall: 0.8810 (0.8706 – 0.8914)
F1 Score: 0.9263 (0.9201 – 0.9323)



- **High Model Performance:** Achieves strong accuracy (~97.6%) and precision (~97.7%), indicating reliable predictions with minimal false positives.
- **Balanced Classification:** F1 score (~92.6%) reflects a good balance between precision and recall, though recall (~88.1%) is slightly lower.



Overall results confirm the hybrid model is robust and well-suited for ECG classification tasks.

- **Stable Training Behavior:** Training and validation loss decrease consistently, showing effective learning without instability.
- **Minimal Overfitting:** Close alignment between training and validation accuracy (~97–98%) indicates good generalization.

THANK YOU!